Multi-Client Delegated Verifiable Blind Quantum Computing

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Abstract

We construct a multi-Client verifiable blind delegated quantum computing (VBDQC) protocol in which M classical Verifier Clients jointly delegate a public quantum circuit acting on their combined private inputs to two quantum-capable Provers sharing entanglement but having no direct communication channel. We begin by modifying the single-Client, entanglement-based CGJV Verifier-on-a-Leash protocol. We then prove simulator-based blind verifiability for the modified protocol using the abstract/constructive cryptography framework. We adapt the DFPR upward-reduction chain to the two-Prover setting, constructing a no-direct-communication simulator architecture. We then extend the CGJV protocol to M Clients using classical multiparty computation techniques, preserving the single-Client guarantees of completeness, soundness, and blindness while additionally guaranteeing input privacy between Clients. Security is analyzed under a threat model that allows a subset of Clients to collude with one but not both of the Provers, which do not communicate even after the protocol execution.

CCS Concepts

• **Theory of computation → Quantum computation theory**; • **Security and privacy → Formal cryptographic methods**; *Protocol composition*; *Multi-party computation.*

Keywords

delegated quantum computing, verifiable blind quantum computing, classical Verifier, two-Prover protocols, multi-party computation, simulator-based security, blind verifiability, entanglement-based protocols

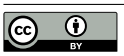
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1 Introduction

In recent years, the advent of cloud-accessible quantum computers has provided access to a wide array of Clients [36]. For quantum cloud computing to fulfill its promise, an important requirement is its security: Clients who outsource computation (i.e. delegate quantum circuits) to a quantum cloud must be assured of the correctness of the computation output and be able to protect the privacy of their inputs. Several cryptographic protocols have been developed to this end. Some protocols require the Client to have limited quantum capabilities [13], while others allow for fully classical Clients either through the use of multiple non-communicating Provers [38] or through the use of cryptographic assumptions on the hardness of certain problems [33], often based on the learning with errors (LWE) problem [37]. Existing V(B)DQC protocols fall into four types: prepare-and-send, receive-and-measure, entanglement-based [24], and more recently developed computationally-secure. They can be further distinguished by the Verifier's computational resources and the trust/rigidity assumptions used.

The Fitzsimons-Kashefi protocol [23] is a prepare-and-send scheme using a measurement-based quantum computing (MBQC) framework with a Verifier requiring only limited quantum power (single-qubit operations). This protocol achieves blindness by applying a quantum one-time-pad [19] to the input angles sent to the Prover, and verifiability through the secret embedding of trap qubits; the protocol is universal, as any circuit can be compiled by beginning with a highly entangled graph state and performing measurements adaptively [23]; a device-independent variant was also proposed [29]. Another prepare-and-send approach is the Broadbent test-or-compute protocol [12] which has at most linear overhead and requires the Verifier to have the ability to prepare single-qubit states. In this case, blindness is achieved through the use of a quantum one-time-pad on the input and output qubits, and verifiability is achieved through the use of indistinguishable test rounds that check the Prover's honesty. A major breakthrough was achieved by lowering the computational resource requirements to purely classical [33] for the Verifier by leveraging the hardness of the learning with errors (LWE) problem [37]. In these Mahadev-style schemes the Verifier uses post-quantum cryptography to generate trapdoor claw-free functions under LWE-hardness to force a quantum Prover's measurements into checkable transcripts, yielding interactive classically verifiable delegation [33]. Alternative classical-Verifier approaches via parallel remote state preparation have also been developed [26], while single-Prover classical-Client delegation has been shown to face fundamental security limitations [6].

In entanglement-based approaches, inspired by Bell's inequality-violating Clauser-Horne-Shimony-Holt (CHSH) game, a fully classical Verifier interacts with two non-communicating Provers possessing shared entangled states [9], a model made necessary in part by fundamental limitations on single-server quantum cryptography [39]. These protocols are typically based on CHSH games, where the remote Provers leverage shared entanglement in response



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to Verifier questions to produce correlations that surpass the maximum classical bound [15]. Rigidity of CHSH games can be used to tie the Provers' behavior to ideal measurements, enabling both verifiability and blindness through self-testing [38, 41]. Ongoing work continues to improve round and communication complexity, fault tolerance, and device-independence [2, 4, 5, 24, 25, 42].

These protocols are all aimed at the single-Client case, where a classical user has a private input and circuit that they wish to delegate to a remote quantum computer to perform. A more useful primitive is one that allows joint computation on private inputs by several Clients (e.g., hospitals participating in research studies cannot freely share confidential patient data, yet there is value in running the study on a federated corpus of data). This problem has been extensively studied on the classical side: multiparty computation (MPC) lets mutually distrustful parties compute a public function $f(x_1, \ldots, x_n)$ on private inputs without revealing anything beyond f's output. Two foundational paradigms underpin most deployed systems. Garbled circuits (Yao) support efficient two-party computation, achieving security in the semi-honest model and, with standard compiler techniques, against malicious adversaries [8, 28, 44]. Secret-sharing-based MPC generalizes to many parties with information-theoretic security under an honest majority assumption [11, 40]. In practice, protocols compose these tools in different ways taking into account efficiency and computational limitations across different threat models, with modern systems routinely handling millions of gates with modest online latency [18, 22].

Recent work has begun to explore the multi-Client quantum case, where multiple Clients with private inputs wish to jointly delegate a public circuit to a remote quantum computer. Unconditionally secure quantum multiparty computation has been studied in both the honest-majority [10] and dishonest-majority [20] settings, as well as for asymmetric Client configurations [30]. The Kashefi-Pappa prepare-and-send protocol requires Clients to have weak quantum capability and does not provide verifiability [31]. Bartusek instead provides verifiability for purely classical Clients with computational security based on a Mahadev-style protocol [7].

A conceptually simple, yet not fully secure, approach would be to implement an *MPC Oracle* on top of a single-classical-Client delegation protocol: the quantum cloud would interact with a single virtual Client whose input is the collection of the M individual inputs of the parties. An MPC protocol would be run between the M parties at each round to compute the "next message" sent to the quantum cloud by the virtual Client. But MPC only guarantees privacy of the inputs against small coalitions of Clients. As shown by Bartusek [7], even a single Client *colluding with one of the quantum servers* can learn information about the other Clients' inputs by exploiting input-dependent aborts. In other words, the server can compute a function that causes Clients to abort based on the value of their input, and the colluding Client can then tell the server whether the other Clients aborted. Bartusek's solution to this problem, which uses an MPC oracle to build a Mahadev-style multi-Client single-server protocol, greatly increases the computational complexity of the protocol for both Clients and servers.

Our contribution. In this paper, we incorporate ideas from classical MPC to modify an entanglement-based V(B)DQC protocol to compute a public quantum circuit (which can be hidden inside a universal circuit with logarithmic overhead) on private inputs from multiple classical Verifiers with information-theoretic security and only linear overhead. Our protocol provides verifiability and input privacy among Clients in a threat model that allows a subset of Clients to collude with one but not both of the Provers, which do not communicate even after the protocol execution.

We apply the MPC Oracle technique to the CGJV "Verifier-on-a-Leash" protocol (with one classical Verifier and two entangled non-communicating Provers) [16, 17]. To avoid input-dependent aborts, we modify CGJV so that test failures and computation outcomes are recorded separately. This in turn allows us to prove the original security properties of completeness, soundness, and blindness via a simulation-based approach in the abstract cryptography framework of Dunjko et al. [21]. As in Bartusek's approach, this is sufficient to prevent input-dependent aborts.

One interpretation of our result is that Provers' entanglement and physical separation naturally prevent input-dependent aborts, trading the complexity of the Mahadev/Bartusek protocol for the stronger assumption of physical separation of the Provers.

2 Background

2.1 Abstract Cryptography & Threat Model

To model our protocol's security, we use the abstract/constructive cryptography framework [34]. This top-down framework first specifies the complete abstract system and introduces lower-level features as needed. A cryptographic protocol π uses a resource $\mathcal{R}$ to approximately construct another, usually ideal, resource $\mathcal{S}$; such constructions can be composed in parallel or sequentially.

A resource $\mathcal{R}$ has a set of interfaces $\mathcal{I} = \{A, B, C\}$, each exposing prescribed functionality to one user. A filter $\perp$ blocks functionality available only to a malicious user. A converter π_i has an outside interface for inputs and outputs and an inside interface connected to a resource; a protocol $\pi = \{\pi_i\}_{i \in \mathcal{I}}$ provides one converter per resource interface. To determine whether π constructs $\mathcal{S}$ within error ϵ, we use a distinguisher $\mathcal{D}$. For an n-interface resource, $\mathcal{D}$ connects to all n interfaces and has one additional interface that outputs a bit. Its distinguishing advantage defines the pseudo-metric

$$d(\mathcal{R}, \mathcal{S}) = \max_{\mathcal{D}} \left| \Pr\left[\mathcal{D}\mathcal{R} = 1\right] - \Pr\left[\mathcal{D}\mathcal{S} = 1\right] \right|, \tag{1}$$

where the maximum is over the distinguisher class permitted by the architecture and threat model. If $d(\mathcal{R}, \mathcal{S}) \leq \epsilon$, then the resources are ϵ-close, written $\mathcal{R} \approx_\epsilon \mathcal{S}$. Composability is controlled by how this advantage grows under composition. We define an ideal resource and show that there exist converters, called simulators σ_i, that make the ideal resource indistinguishable from the real resource and protocol:

$$d\left(\pi\mathcal{R}, \mathcal{S}\sigma\right) \leq \epsilon. \tag{2}$$

Correctness is defined by filters that restrict every interface to its honest functionality:

$$d\left(\pi\mathcal{R}, \mathcal{S} \perp\right) \leq \epsilon. \tag{3}$$

We are bound by the same restriction as the original CGJV protocol that assumes no communication between the Provers, even after the protocol is complete, which is required for blindness. This assumption is a strong limit on the practicality of this approach.

One avenue to potentially remedy this would be to incorporate an (anonymous) entanglement routing protocol [1] that makes it difficult for two Provers on a large network to find each other, but we leave the possibility of such extensions for future work. Additionally, while we demonstrate soundness and completeness for our modified protocols without requiring one of the two Provers to be honest, for the application of the Dunjko-Fitzsimons-Portmann-Renner (DFPR) reduction chain we introduce the simplifying assumption not present in the CGJV protocol of at least one honest Prover. The motivation for this assumption comes from wanting to incorporate multiparty functionality in a simulation-based security manner using the DFPR reduction. We initially attempted to modify the DFPR proof chain. However, constructing a distinguisher and simulators while restricting the Provers from communicating introduced additional complications that we were unable to resolve, so we leave such an extension for future work. We do however consider the security-relevant case in which a malicious Prover colludes with a subset of Clients, a case omitted in some previous work [31]. Specifically,

$$\text{Clients: } \mathcal{C} \equiv \{C_1, C_2, \ldots, C_M\}, \tag{4}$$

$$\text{Provers: } \mathcal{P} \equiv \{PV, PP\}, \tag{5}$$

$$\text{Colluding subset: } \mathcal{C}' \subseteq \mathcal{C}, \text{ such that } |\mathcal{C}'| \leq t. \tag{6}$$

$$\text{Total coalition: } \mathcal{A} = \mathcal{C}' \cup \{PV\}, \text{ or } \mathcal{C}' \cup \{PP\}. \tag{7}$$

To retain the no-communication assumption, a colluding Client subset may communicate with either PV or PP, but not both. Our assumption of at least one honest Prover also excludes the case of two disjoint collusion subsets containing one Prover each. While our protocol, like other entanglement-based protocols requiring two independent quantum computers with shared entanglement and no-communication, is impractical in the near-term, there is growing hardware investment in quantum networks [45] that could eventually be used to implement complex quantum multiparty networking protocols such as ours that offer security benefits without relying on computational assumptions.

We are interested in the case of purely classical Clients that share classical channels with one another and the Provers, which is closer to current practical reality of classical devices communicating with cloud-accessible quantum devices. For later technical simplicity we represent classical registers as quantum registers diagonal in a fixed computational basis. Classical communication is enforced by the complete dephasing map which kills all off-diagonal elements

$$\Delta_C(\rho) = \sum_c |c\rangle \langle c|_C \rho |c\rangle \langle c|_C . \tag{8}$$

Thus an arbitrary quantum round map $\mathcal{F}_i$ is operationally equivalent at a classical interface to

$$\hat{\mathcal{F}}_i = (\Delta_C \otimes \mathrm{id}_R) \circ \mathcal{F}_i \circ (\Delta_C \otimes \mathrm{id}_R). \tag{9}$$

Since dephasing and all later classical processing are Completely-Positive Trace-Preserving (CPTP) maps, trace distance is contractive under these operations. For simplicity, we therefore treat classical channels as a special case of quantum channels and use the quantum-channel formalism from this point forward.

Protocol	Provers	Rounds	Resources
RUV 2012 [38]	2	poly(n)	$\geq g^{8192}$
McKague 2013 [35]	poly(n)	poly(n)	$\geq 2^{153} g^{22}$
GKW 2015 [25]	2	poly(n)	$\geq g^{2048}$
HPDF 2015 [29]	poly(n)	poly(n)	$O(g^4 log\ g)$
GV* 2019 [27]	1	$O(g^4)$	$O(g^4)$
CGJV Leash 2024 [17]	2	O(depth)	$O(glog\ g)$

Table 1: A comparison of various VBDQC protocols that involve a purely classical Verifier interacting with remote non-communicating but entangled quantum Provers, while the Gheorghiu-Vidick protocol is distinct from the others as it uses only a single quantum server and computational security. Here n is the number of qubits needed and g is the number of gates in the target circuit, and resources refers to the requirements to implement the protocol including the number of gates, the number of prior entangled qubits, and the number of bits communicated throughout the protocol. This table is based on Table 1 included in [17].

2.2 CGJV Leash Protocol

We begin with an overview of the original CGJV Verifier-on-a-Leash protocol [17], which takes the form of an interactive non-local quantum game, with a fully classical Verifier V, and two quantum Provers that share entanglement PV and PP. This protocol contains two branches, one is the EPR version of Broadbent's test-or-compute VBDQC protocol [12] while the other is a novel rigidity branch that consists of many different sub-branches, one of which is the CHSH game. The CHSH game involves two entangled Provers that can attain the optimal quantum winning probability $\cos^2(\pi/8)$, exceeding the classical bound of 3/4 [14, 15, 43]. Rigidity strengthens this gap into a self-test where a near-optimal strategy is, up to local isometries, close to Bell pairs measured with the ideal observables [5, 42]. There are many extensions of the original CHSH game and these extensions have been incorporated in a range of VBDQC protocols, some of which are included in Table 1, where, with the exception of the Gheorghiu-Vidick (GV) computationally secure protocol, the included protocols do not rely on computational hardness assumptions. As illustrated, the CGJV Leash protocol requires the least overhead and this is why it is our chosen protocol for multiparty extension.

The CGJV Leash protocol uses CHSH games in conjunction with more sophisticated tests such as the magic square game [3] to create their rigidity branch as shown in Fig. 1. The rigidity branch certifies the multi-qubit measurements required for Broadbent's EPR protocol (summarized in Section 3.1) which forms the delegation game branch of the CGJV Leash protocol. The CGJV protocol is a two-Prover entanglement-based VBDQC protocol with a classical Verifier V and non-communicating Provers PV and PP, where the rigidity branch certifies that PV behaves like V_{EPR} in Broadbent's EPR protocol. We describe an honest execution before recalling the sequential wrapper used to amplify the completeness-soundness gap.

The Provers PV and PP share $m = \Theta(n + t)$ EPR pairs, where n is the input size of the universal circuit Q that has been broken down into layers such that each layer contains at most 1 T-gate per qubit

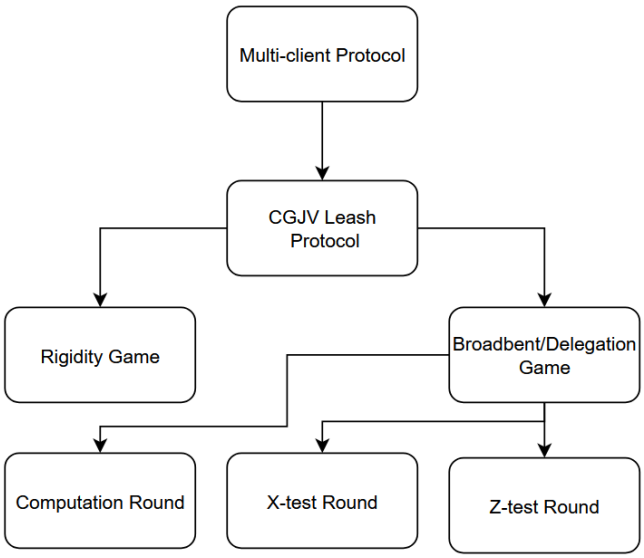


Figure 1: The structure of the CGJV Leash Protocol and its interface with our intended MPC protocol. It has a rigidity branch, which self-tests the Provers and a delegation branch, which performs the desired computation.

and $t = t_1 + \cdots + t_d$ is the total number of T gates across the d T-gate layers. This is the required circuit structure for Broadbent's EPR protocol [12]. Its security is based on the fact that the Prover, (in this case PP) cannot distinguish between computation rounds and test rounds and is forced to behave honestly or risk being caught in one of the test rounds. Broadbent's protocol requires an interactive step between Verifier and Prover to implement T-gates using prepared ancillary qubits that act as a resource for T-gate implementation for a particular round type. These T-gadgets are deterministic in the test rounds which allows the Verifier to treat them as traps to catch a cheating Prover.

In CGJV Leash the Verifier V begins by flipping a biased coin, executing the rigidity game with probability p_r and the delegation game with probability $1 - p_r$. In the rigidity branch, V plays a sequential version of the self-testing game named RIGID(Σ, m) with both Provers. Here $\Sigma = \{X, Y, Z, F, G\}$ is the alphabet consisting of instructions to measure in the various X, Y, Z Pauli bases as well as the $F = (-X + Y)/\sqrt{2}$ and $G = (X + Y)/\sqrt{2}$ bases and m is the string length. The RIGID(Σ, m) game is structured in a way that makes (at least a subset of it) indistinguishable from the delegation game for PV, by sampling over the same messages and breaking the exchanges down to match the input + layer-by-layer T-gadget structure of the delegation game. We refer the readers to the original text for the complete definition of RIGID(Σ, m).

The Verifier V's view consists of the question pair (W, W'), where $W \in \Sigma^m$ is sampled according to the question distribution of RIGID(Σ, m) and sent to PV and PP, respectively, and the various classical answer strings $(\vec{e}, \vec{e}\,')$ the Provers return, which correspond to measurement outcomes in the honest case. V accepts if and only if the answers satisfy the winning condition of RIGID(Σ, m). Sequential play, in which the parties exchange certain subsets of W/W' questions and answers in sequence while retaining the same acceptance condition, only reduces the Provers' ability to cheat. In the delegation branch, from V's point of view, the following steps are performed in an honest execution:

(1) V samples $W \in_R \Sigma^m$ uniformly at random and partitions $\{1, \ldots, m\}$ into subsets A and $B_1, \ldots, B_d$ which will be used for the circuit input and for the T-gadgets in each circuit layer, respectively, such that W_A contains at least n copies of each symbol in Σ and each W_{B_ℓ} contains at least t_ℓ copies of each symbol.
(2) V sends (A, W_A) to PV, who measures the indicated halves of his EPR pairs in the bases specified by W_A and returns outcomes $\vec{e}_A$. V then sequentially sends pairs of indices and measurement bases $(B_1, W_{B_1}), \ldots, (B_d, W_{B_d})$ for each circuit layer ℓ. Each time V receives outcomes $\vec{e}_{B_\ell}$. Since PV honestly measures halves of EPR pairs, each outcome bit in V's view is uniformly distributed.
(3) V selects a run type $r \in \{\text{computation}, \text{X-test}, \text{Z-test}\}$ uniformly at random and, as a function of r and W, selects index sets $N \subseteq A$ with $|N| = n$ (the input qubits) and $T_\ell = T_\ell^0 \cup T_\ell^1 \subseteq B_\ell$ with $|T_\ell| = t_\ell$ (the ancillary T gate qubits). The subsets T^0 and T^1 are needed for different round types. V then sends the sets N, T_ℓ^0, T_ℓ^1 (but not W or $\vec{e}$) to PP. Note that the selection of input qubits and the measurement outcomes from PV and round type determine the initial quantum one-time pad keys.
(4) For each layer $\ell = 1, \ldots, d$, V performs the key updates for the Clifford gates and receives measurement outcomes $\vec{c}_{T_\ell} = \{c_i\}_{i \in T_\ell}$ from PP. V checks the traps in test rounds, rejecting if $c_i \neq a_j \oplus e_i$ for any $i \in T_\ell^0$ in an X-test run or $i \in T_\ell^1$ in a Z-test run. V then computes the corrections z_i (in a computation run $z_i = a_j + \mathbf{1}_{W_i = F} + c_i$, while in test runs z_i is either uniformly random or determined by W_i) and sends $\vec{z}_{T_\ell}$ to PP, then performs the T-gadget key updates.
(5) V receives the output bit c_f from PP. In a computation or X-test run V rejects if $c_f \oplus a_f \neq 0$, where a_f is the final X-key on the output wire, and accepts otherwise.

In the honest case, V's view in the delegation branch consists of her own random sampling $(r, W, N, T_\ell^0, T_\ell^1)$, the uniformly random outcomes $\vec{e}$ from PV, the outcome bits $\vec{c}$ and c_f from PP, and the corrections $\vec{z}$ and pad keys she computes from these. The combined action of V and honest PV is exactly that of V_{EPR} in Broadbent's EPR protocol, with W chosen at random and $\vec{z}$ chosen accordingly rather than the reverse, so the joint distribution of PP's interaction is identical to that of the Prover in the EPR protocol.

CGJV prove completeness, soundness, and blindness for this protocol. They prove a constant completeness-soundness gap, which they then magnify through a repeated sequential wrapper that we introduce shortly. Their completeness proof reduces to the Broadbent EPR completeness. Their soundness proof first reduces delegation attacks with honest PV to attacks on Broadbent's EPR protocol, then uses rigidity to replace near-optimal arbitrary Provers by a pair with honest PV. Lastly, a reordering argument establishes blindness for each Prover's local view.

2.2.1 Sequential Wrapper. The single-shot Leash protocol guarantees only a small constant completeness-soundness gap $\Delta =$

$p_{comp} - p_{sound}$, and its accept/reject decision cannot by itself be interpreted as the outcome of the computation branch. That is simply interpreting acceptance as a "YES" outcome and rejection as a "NO" outcome is insufficient as noted in CGJV. Although cheating Provers have a low probability of making V accept a "NO" instance, they can always make V reject a "YES" instance by deliberately failing the tests or simply refusing to respond. CGJV address both issues with a sequential wrapper $Seq(P, c, \Delta, K)$, where P is the Leash protocol with completeness $c = p_{comp}$ and gap Δ, and K is a security parameter [17]. In the sequential wrapper, V runs K copies of P in sequence on input $(Q, \vec{x})$, and then K copies in sequence on input $(Q', \vec{x})$, where Q' is the circuit obtained by appending an X gate to the output wire of Q. If Q decides a function f with bounded error $1/3$, then Q' decides $1 - f$ with the same error, so on any instance exactly one of the two batches should accept at a high rate. Let $\vec{o}, \vec{\tilde{o}} \in \{0,1\}^K$ record which copies accepted on $(Q, \vec{x})$ and $(Q', \vec{x})$ respectively. V applies the accept-reject dual criteria to the Hamming weights of these strings against the threshold $(c - \frac{\Delta}{2})K$, set midway between the completeness and soundness rates. She then outputs 0 (accept-reject) if $wt(\vec{o}) \geq (c - \frac{\Delta}{2})K$ and $wt(\vec{\tilde{o}}) < (c - \frac{\Delta}{2})K$, outputs 1 (reject-accept) if $wt(\vec{o}) < (c - \frac{\Delta}{2})K$ and $wt(\vec{\tilde{o}}) \geq (c - \frac{\Delta}{2})K$, and aborts on the remaining accept-accept and reject-reject patterns, which occur with low probability in the honest provers case. Note that we have flipped the output convention here for later clarity. The dual criteria ensure that Provers who deliberately fail can only force an abort, never the wrong outcome ("YES" or "NO") except with low probability as to make V output the wrong bit they would have to pass the threshold on the batch that should reject, which contradicts soundness. Concretely, for honest Provers V outputs $f(\vec{x})$ with probability at least $1 - 2e^{-\Delta^2 K/2}$, while for any cheating Provers the probability that V outputs $1 - f(\vec{x})$ is at most $e^{-\Delta^2 K/8}$. So V either aborts or obtains the correct outcome except with probability exponentially small in K. One subtlety in the soundness proof is that, because the Leash protocol is blind, Provers in later copies cannot simply be treated as black boxes conditioned on earlier transcripts, the argument in CGJV instead uses the observation that the Leash protocol remains sound even when $\vec{x}$ is revealed to the Provers, together with an Azuma-Hoeffding martingale bound over the sequential runs. Since the gap Δ is constant, it can be easily amplified through repetition.

2.3 DFPR Local Reduction Chain

The CGJV Leash protocol does not prove blindness in a simulatable and composable way, it leaves this as an open problem [17]. To achieve this stronger notion of blindness, which is necessary for later multiparty inclusion [7], we follow the DFPR strategy. This strategy reduces simulatable blind-verifiability to local properties that can be proved directly [21, 32]. In Fig. 2 we illustrate the original DFPR proof structure in solid lines. We have also included our contribution where we modify the CGJV Leash protocol in order to be able to prove Independent Local-Verifiability which is necessary for us to construct our ideal functionality that can be composed with a classical multiparty protocol.

In DFPR, the ideal resource $\mathcal{S}^{blind}_{verify}$ accepts an arbitrary state $|\psi\rangle$ and returns either the correct output of a public circuit $U|\psi\rangle$ or an orthogonal error state $|err\rangle$. A protocol ϵ-close to this resource

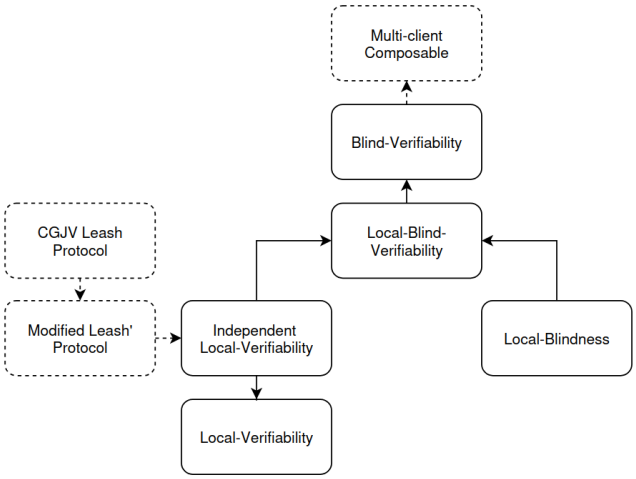


Figure 2: The original DFPR proof chain is illustrated in solid lines with the relevant direction used in our result, where both Local-Blindness and Independent Local-Verifiability are necessary to prove Local-Blind-Verifiability. In dashed lines we include our contribution where we modify the CGJV Leash protocol in order to prove Independent Local-Verifiability and we compose the ideal functionality with a classical multiparty computation protocol.

is ϵ-blind-verifiable. DFPR relate local blindness, local verifiability, independent local-verifiability, and local blind-verifiability through

$$\epsilon_{ind} - \text{independent } \epsilon_{ver} - \text{local-verifiability} + \epsilon_{bl} - \text{local-blindness} \implies \delta - \text{local-blind-verifiability}. \quad (10)$$

DFPR Lemma 6.6 and Theorem 6.7 give $\delta = 2\sqrt{2\epsilon_{ver}} + \epsilon_{bl} + \epsilon_{ind}$, while their Theorem 5.3 gives

$$\delta\text{-local-blind-verifiability} \implies 2\delta\text{-blind-verifiability}. \quad (11)$$

The local properties are defined as follows:

Definition 2.1 (Local-Blindness). *A DQC protocol provides ϵ_{bl}-local-blindness if, for all adversarial behaviors $\{\mathcal{F}_i\}$, there exists a CPTP map $\mathcal{F} : \mathcal{L}(\mathcal{H}_B) \to \mathcal{L}(\mathcal{H}_B)$ such that*

$$tr_A \circ \mathcal{P}_{AB} \approx_\epsilon \mathcal{F} \circ tr_A. \quad (12)$$

Here $\mathcal{L}(\mathcal{H})$ is the set of all linear operators on Hilbert space $\mathcal{H}$ and $\mathcal{P}_{AB}$ is the map corresponding to the protocol run with honest Alice and adversarial Bob, and tr_A is the partial trace over Alice's system.

Definition 2.2 (Local-Verifiability). *A DQC protocol provides ϵ_{ver}-local-verifiability if, for all adversarial behaviors $\{\mathcal{F}_i\}$, and all initial states $\psi_{A,R_1} \otimes \psi_{R_2,B}$ there exists a probability p^ψ such that*

$$\rho^\psi_{AR_1} \approx_\epsilon p^\psi (U \otimes id_{R_1})(\psi_{AR_1}) + (1 - p^\psi) |err\rangle\langle err| \otimes \psi_{R_1}, \quad (13)$$

where $\rho^\psi_{AR_1}$ is the final state of the protocol for Alice and the first reference system, $(U \otimes id_{R_1})(\psi_{AR_1})$ is the correct output state, $|err\rangle$ is an orthogonal error state, and ψ_{R_1} is the reduced state of R_1.

Definition 2.3 (Independent Local-Verifiability). *A DQC protocol provides ϵ_{ind}-independent ϵ_{ver}-local-verifiability if it provides*

ϵ_{ver}-local-verifiability and, for every adversarial map

$$\mathcal{F}_i : \mathcal{L}(\mathcal{H}_{CB}) \longrightarrow \mathcal{L}(\mathcal{H}_{CB}),$$

there is an alternative map

$$\mathcal{F}'_i : \mathcal{L}(\mathcal{H}_{CB\bar{B}}) \longrightarrow \mathcal{L}(\mathcal{H}_{CB\bar{B}})$$

such that

$$tr_A \circ \mathcal{Q}_{A\bar{B}} \circ \mathcal{P}_{AB} \approx_\epsilon tr_A \circ \mathcal{P}'_{AB\bar{B}}. \tag{14}$$

Here $\mathcal{P}_{AB}$ is the map corresponding to the protocol run with honest Alice and adversarial Bob, and $\mathcal{Q}_{A\bar{B}}$ is a map that copies the accept/reject decision of Alice to an additional system $\bar{B}$.

Finally, local blind-verifiability is defined as follows:

Definition 2.4 (Local-Blind-Verifiability). *A DQC protocol provides ϵ-local-blind-verifiability if, for all adversarial behaviors $\{\mathcal{F}_i\}$, there exist two completely positive trace-nonincreasing maps $\mathcal{F}_B^{ok}, \mathcal{F}_B^{err}$ such that*

$$\mathcal{P}_{AB} \approx_\epsilon \mathcal{U}_A \otimes \mathcal{F}_B^{ok} + \mathcal{E}_A^{err} \otimes \mathcal{F}_B^{err}. \tag{15}$$

Here $\mathcal{P}_{AB}$ is the map corresponding to the protocol run with honest Alice and adversarial Bob, $\mathcal{U}_A$ is the map corresponding to the correct unitary operation on Alice's system, and $\mathcal{E}_A^{err}$ is the map corresponding to the error state on Alice's system.

Under our threat model with one honest Prover, we can absorb the honest Prover into the real resource and reduce the two-Prover protocol to a single-adversary DQC protocol for which we can use the DFPR reduction directly. What remains then is to prove the local properties directly, that is proving ϵ_{ind}-independent ϵ_{ver}-local-verifiability and ϵ_{bl}-local-blindness for both the case of honest *PV* and for the case of honest *PP*.

3 Single-Client Modified CGJV Protocol (Leash')

We construct a modified CGJV Leash protocol that we call Leash' that composes securely with classical MPC under our threat model. In the original CGJV Leash protocol, a failed rigidity or delegation test and an honest computation outcome both contribute to one accept/reject decision. A colluding Prover that learns whether the MPC aborts could exploit this input-dependent signal, as described by Bartusek and DFPR [7, 21]. In order for us to use the DFPR reduction that would prevent this issue we are required to prove independent local-verifiability where the Verifier's valid/error decision must be independent of the input. Intuitively this can be understood as showing whether the Verifier aborts the protocol is independent of the input so any collusion that results in the Prover learning whether the Verifier aborted does not leak any extra information. To prove this property we have to adapt the CGJV leash protocol, and its Broadbent subprotocol to avoid any connection between test outcomes that could result in the Verifier aborting, and computational outcomes that are input dependent. We therefore separate test outcomes from computation outcomes while preserving completeness, soundness, and blindness. Additionally, we prove soundness in the same manner as CGJV, without assuming that one of the Provers is honest, which we assume only later when we go on to using the DFPR reduction.

3.1 Modified Broadbent EPR Protocol

We start by first modifying Broadbent's EPR protocol, which forms a branch of the CGJV Leash protocol as shown in Fig. 1. Broadbent's linear-overhead verification protocol [12] delegates a Clifford+T circuit to one quantum Prover while a Verifier hides the input with a quantum one-time pad and chooses uniformly among indistinguishable computation, X-test, and Z-test runs. Its EPR formulation replaces the Verifier's prepared states with measurements of halves of shared EPR pairs. As local unitaries of the Prover and Verifier commute, this formulation retains the prepared-state protocol's completeness (8/9) and soundness bounds (7/9). The one-time pad and uniformly masked correction messages also make the Prover's view independent of the input except for its permitted size leakage [12]. The key idea behind Broadbent's protocol is that the Verifier can deterministically update the pads through all of the Clifford gates without any interaction needed; this is called "tracking the Pauli frame". As the Clifford set is not universal, an additional interactive T-gate gadget is introduced that preserves blindness but uses an ancillary qubit. The Verifier's key updates for the T-gate gadget are dependent on a local measurement performed by the Prover as well as the run type. The exact update rules are given in Broadbent's original paper [12].

We modify the output set of a single Broadbent EPR round while maintaining completeness and soundness. We then use the modified output set to separate the test and computation outcomes in CGJV Leash. Throughout, the desired circuit is embedded in a universal circuit U, which preserves circuit privacy with only logarithmic overhead in size and depth. The resulting promise problem is to determine

$$f(x) = \begin{cases} 0, & ||\Pi_0 U\,|x\rangle\,||^2 \geq 2/3, \\ 1, & ||\Pi_0 U\,|x\rangle\,||^2 \leq 1/3, \end{cases} \tag{16}$$

Which specifies the allowed inputs used from here onward. Thus, in an honest computation round, the Verifier receives the correct output bit Y with probability at least $2/3$:

$$\Pr\big[Y = f(x) | \text{computation}\big] \geq 2/3. \tag{17}$$

We extend the output alphabet to $Y \in \{0, 1, \text{err}, \text{ok}\}$, defined by

$$Y = \begin{cases} 0, & \text{computation run and depadded output bit } = 0, \\ 1, & \text{computation run and depadded output bit } = 1, \\ \text{err}, & \text{failed test run or broken transcript,} \\ \text{ok}, & \text{test run and passed test.} \end{cases} \tag{18}$$

Note that the original single-round output can easily be recovered by the map

$$A(Y) = 1\big[Y = 0\big] + 1\big[Y = \text{ok}\big]. \tag{19}$$

The honest outcome probabilities of the modified protocol are

$$\Pr\big[Y = f(x) | \text{computation run}\big] \geq 2/3, \tag{20}$$

$$\Pr\big[Y = \text{err} | \text{test run}\big] = 0. \tag{21}$$

Broadbent reduces an arbitrary attack to a convex combination of Pauli attacks using the Pauli twirl lemma. For a circuit on n data qubits with t T-gate gadgets, the final attack acts on

$$m_B = 2t + n \tag{22}$$

measured and data registers. We define the l-qubit Pauli gates as $\mathcal{P}_l = \{I, X, Y, Z\}^{\otimes l}$. Broadbent makes the distinction between non-benign Paulis $\mathcal{N}_l \subset \mathcal{P}_l$ which are those in $\mathcal{P}_l$ that have either an X or a Y gate on at least one measured qubit, and benign Paulis which do not and therefore have no effect on measurement outcome probabilities in the Z basis. As any arbitrary CPTP map can be represented in Kraus operator E_k form, we can then expand the Kraus operator in the Pauli basis summing over $P \in \mathcal{P}_l$

$$E_k = \sum_P \alpha_{k,P} P. \tag{23}$$

Broadbent then averages over the Verifier's hidden keys which gives $w_P \geq 0$ the Pauli-mixture weights after twirling, normalized so that $\sum_P w_P = 1$. Then we can define the total non-benign weight

$$\nu = \sum_{P \in \mathcal{N}_{m_B}} w_P. \tag{24}$$

Then using Broadbent's analysis our test round error under arbitrary deviation becomes

$$\Pr\big[Y = \text{err}|\text{test}\big] \geq \nu/2. \tag{25}$$

Assuming the worst case, that every non-benign Pauli attack causes an incorrect measurement outcome, and the honest or benign attack error is at most 1/3, the single run wrong output probability is given by

$$\Pr\big[Y = 1 - f(x)|\text{comp}\big] \leq \frac{1-\nu}{3} + \nu. \tag{26}$$

Plugging in the test run error bound gives the following single round soundness relation

$$\Pr\big[Y = 1 - f(x)|\text{comp}\big] \leq \frac{1}{3} + \frac{4}{3}\Pr\big[Y = \text{err}|\text{test}\big]. \tag{27}$$

Thus a malicious Prover can increase the probability of an incorrect valid output in a computation run only by increasing the probability of failing a test run. Next we will incorporate this modified Broadbent protocol into a modified Leash protocol that we will adapt, constructing a new sequential wrapper, before proving its security.

3.2 Modified CGJV Protocol

For our Leash' protocol, we assume that all rounds are completed without early termination and that the timing of all messages follows a fixed schedule. We also assume that any broken or incomplete transcripts cause the Verifier to abort at the end of the protocol, and that m is sufficiently large such that a uniformly sampled $W \in \Sigma^m$ contains at least $n + t$ copies of each symbol, and we omit negligibly small sampling corrections going forward. Leash' uses the modified Broadbent protocol in CGJV's delegation branch, played with probability $1 - p_r$, and CGJV's rigidity branch, played with probability p_r that we define shortly. A single round outputs $y_i \in \{0, 1, \text{ok}, \text{err}\}$, a computation round returns to the Verifier c_f who removes their pads and outputs a bit, while the rigidity branch returns ok or err. Only the Verifier's final post-processing changes, so the message distribution is identical to that of the original Leash protocol and the Provers cannot distinguish the two. We leave optimizing parameter choices for future work.

The CGJV Leash protocol's rigidity branch RIGID(Σ, m) consists of a large number of sampled subtests, only one of which (the CHSH subtest) has an honest Prover error probability. We denote the whole rigidity branch honest error probability $R_H = \Pr[y_i = \text{err}|\text{rigid, honest}]$, while the general rigidity error probability for an arbitrary strategy we label $R = \Pr[y_i = \text{err}|\text{rigid}]$. From CGJV's rigidity theorem and Lemma 4.4 [17] we use $\delta(R)$, a non-decreasing upper bound on the statistical transcript distance of the delegation branch between an arbitrary Prover pair strategy with rigidity error R and a transformed Prover pair strategy such that PV is honest and PP is arbitrary and that $\delta(R) \to 0$ as $R \to 0$. We also introduce a rigidity error cutoff, R_0, such that $\delta(R_0) \leq 1/120$, where $1/120$ is a convenient non-optimal parameter. As any arbitrary Prover strategy has a rigidity error R, we can think of R_0 as a boundary where any strategy with $R \leq R_0$ is close enough that we can transform it to the honest PV version up to our desired distance tolerance $1/120$. For a fixed input x and an arbitrary single-round Prover strategy, we define $q = \Pr[y_i = \text{err}|X/Z\text{-test}]$ and $\omega = \Pr[y_i = 1 - f(x)|\text{comp}]$. Let q' and ω' be the corresponding probabilities for the transformed Provers from CGJV's rigidity theorem and Lemma 4.4. Since computation and combined X/Z-test runs have probabilities $1/3$ and $2/3$, respectively,

$$|\omega - \omega'| \leq 3\delta(R), \tag{28}$$

$$|q - q'| \leq \frac{3}{2}\delta(R). \tag{29}$$

The transformed strategy has honest PV, so the modified Broadbent tradeoff gives $\omega' \leq 1/3 + 4q'/3$. Then

$$\begin{aligned} \omega &\leq \omega' + 3\delta(R) \\ &\leq \frac{1}{3} + \frac{4}{3}q' + 3\delta(R) \\ &\leq \frac{1}{3} + \frac{4}{3}q + 5\delta(R). \end{aligned} \tag{30}$$

If $R \leq R_0$, that is the case where we are close enough to use our honest PV transformed strategy replacement, monotonicity gives $5\delta(R) \leq 1/24$. Otherwise if $R > R_0$, we instead use the trivial bound $\omega \leq 1$. Combining these cases gives the single-round bound

$$\omega \leq \frac{1}{3} + \frac{4}{3}q + \frac{1}{24} + \frac{R}{R_0}. \tag{31}$$

For completeness, we choose m sufficiently large that $R_H \leq \frac{R_0}{64}$. We now repeat Leash' for $K \geq 1$ rounds with rigidity-branch probability p_r. For round i, let $g_i \in \{\text{rig}, \text{del}\}$ be its branch label, and let $r_i \in \{0, 1, 2\}$ denote computation, X-test, and Z-test runs, respectively, that are sampled uniformly. For the sequential wrapper we introduce computational outcome counts M_0, M_1 as well as test and rigidity error counts $E_{\text{test}}, E_{\text{rigid}}$ such that

$$M_b = \sum_{i=1}^{K} \mathbf{1}[y_i = b], \quad b \in \{0, 1\}, \tag{32}$$

$$M_c = M_0 + M_1, \tag{33}$$

$$E_{\text{test}} = \sum_{i=1}^{K} \mathbf{1}[y_i = \text{err},\ g_i = \text{del},\ r_i \in \{1, 2\}], \tag{34}$$

$$E_{\text{rigid}} = \sum_{i=1}^{K} \mathbf{1}[y_i = \text{err},\ g_i = \text{rig}]. \tag{35}$$

Like in CGJV, we introduce a sequential wrapper to amplify the completeness-soundness gap. Our modified sequential wrapper with K repetitions is defined as

$$W_L(K) = \begin{cases} \text{err}, & E_{\text{test}} > 0, \\ \text{err}, & E_{\text{rigid}} > p_r R_0 K/32, \\ 0, & M_0 > M_1, \\ 1, & M_1 > M_0, \\ b_L, & M_0 = M_1, \end{cases} \tag{36}$$

where b_L is a uniformly random sampled tie-break bit. Note that the valid/error decision in our new wrapper depends on transcript validity and the two error counters, not on M_0 or M_1, which we will need later to show independent local verifiability. The following results establish completeness and soundness for these non-optimized parameters.

Theorem 3.1 (Sequential Leash' Completeness). *Let $p_r \in (0,1)$, $K \geq 1$, and choose the rigidity instance size so that $R_H \leq R_0/64$. For honest behavior and any input x, the probability that the sequential Leash' wrapper $W_L(K)$ outputs the correct result is at least*

$$\Pr[W_L(K) = f(x) | \text{honest}] \geq 1 - \left(1 - \frac{1-p_r}{3} + \frac{1-p_r}{3} e^{-1/18}\right)^K - e^{-p_r^2 R_0^2 K/2048}. \tag{37}$$

Proof. Under honest behavior, the expected number of honest rigidity errors satisfies

$$\mathbb{E}[E_{\text{rigid}} | \text{honest}] = p_r R_H K \leq \frac{p_r R_0 K}{64}. \tag{38}$$

The rigidity threshold $p_r R_0 K/32$ is twice this upper bound. Since the honest rigidity-failure indicators are independent and lie in $[0,1]$, Hoeffding's inequality gives

$$\Pr\left[E_{\text{rigid}} > \frac{p_r R_0 K}{32} | \text{ honest}\right] \leq \exp\left(-\frac{2(p_r R_0 K/64)^2}{K}\right) = e^{-p_r^2 R_0^2 K/2048}. \tag{39}$$

We next bound the probability of an incorrect computation majority. Condition on $M_c = j$ and let H_j be the number of correct outputs among these j honest rounds. For $j \geq 1$, each output is independently correct with probability at least $2/3$. A wrong majority, including a wrong random tie-break, requires $H_j \leq j/2$, so Hoeffding's inequality gives

$$\Pr\left[H_j \leq j/2 | M_c = j, \text{honest}\right] \leq \exp\left(-2j[2/3 - 1/2]^2\right) = e^{-j/18}. \tag{40}$$

Averaging over M_c therefore gives

$$\begin{aligned} \Pr\left[H_{M_c} \leq M_c/2 | \text{honest}\right] &\leq \sum_{j=0}^{K} e^{-j/18} \cdot \binom{K}{j} \left(\frac{1-p_r}{3}\right)^j \left(1 - \frac{1-p_r}{3}\right)^{K-j} \\ &= \left(1 - \frac{1-p_r}{3} + \frac{1-p_r}{3} e^{-1/18}\right)^K. \end{aligned} \tag{41}$$

An honest incorrect final output requires either excessive rigidity errors or a wrong computation majority (or tie-break). Simply taking the union bound over these events then proves completeness. □

Theorem 3.2 (Sequential Leash' Soundness). *Let $p_r \in (0,1)$, $K \geq 1$, and let $R_0 \in (0,1]$ be the rigidity cutoff. For any input x and every adaptive pair of Prover strategies, the probability that the sequential Leash' wrapper outputs the incorrect result is at most*

$$\Pr[W_L(K) = 1 - f(x)] \leq e^{-(1-p_r)^2 K/512\gamma^2} \tag{42}$$

where

$$\lambda = \frac{1-p_r}{3R_0 p_r}, \quad \gamma = 1/2 + \max\{1, \lambda\}. \tag{43}$$

Proof. For a fixed input x and an arbitrary adaptive pair of Prover strategies, let $\mathcal{T}_i$ denote the classical transcript through round i, including the branch and run labels, messages, round outputs, and Verifier randomness used so far. For every possible transcript, we define

$$R_i = \Pr[y_i = \text{err} | g_i = \text{rig}, \mathcal{T}_{i-1}], \tag{44}$$

$$q_i = \Pr[y_i = \text{err} | g_i = \text{del}, r_i \in \{1,2\}, \mathcal{T}_{i-1}], \tag{45}$$

$$\omega_i = \Pr[y_i = 1 - f(x) | g_i = \text{del}, r_i = 0, \mathcal{T}_{i-1}]. \tag{46}$$

The single-round tradeoff (31) applies to each conditioned strategy

$$\omega_i \leq \frac{1}{3} + \frac{4}{3} q_i + \frac{1}{24} + \frac{R_i}{R_0}. \tag{47}$$

For each round i, we define the indicators

$$w_i = \mathbf{1}[g_i = \text{del},\ r_i = 0,\ y_i = 1 - f(x)], \tag{48}$$

$$k_i = \mathbf{1}[g_i = \text{del},\ r_i = 0], \tag{49}$$

$$d_i = \mathbf{1}[g_i = \text{del},\ r_i \in \{1,2\},\ y_i = \text{err}], \tag{50}$$

$$h_i = \mathbf{1}[g_i = \text{rig},\ y_i = \text{err}]. \tag{51}$$

The conditional expectations are

$$\mathbb{E}[w_i | \mathcal{T}_{i-1}] = \frac{1-p_r}{3} \omega_i, \tag{52}$$

$$\mathbb{E}[k_i | \mathcal{T}_{i-1}] = \frac{1-p_r}{3}, \tag{53}$$

$$\mathbb{E}[d_i | \mathcal{T}_{i-1}] = \frac{2(1-p_r)}{3} q_i, \tag{54}$$

$$\mathbb{E}[h_i | \mathcal{T}_{i-1}] = p_r R_i. \tag{55}$$

Using (47), we obtain

$$\begin{aligned} \mathbb{E}[w_i - d_i | \mathcal{T}_{i-1}] &\leq \frac{1-p_r}{8} - \frac{2(1-p_r)}{9} q_i + \frac{1-p_r}{3R_0} R_i \\ &\leq \frac{1-p_r}{8} + \frac{1-p_r}{3R_0} R_i. \end{aligned} \tag{56}$$

We now introduce a weighting term $\lambda = (1-p_r)/(3R_0 p_r)$ on the rigidity errors to construct our supermartingale to allow us to cancel the rigidity-dependent term by subtracting λh_i

$$\mathbb{E}[w_i - k_i/2 - d_i - \lambda h_i + (1-p_r)/24 | \mathcal{T}_{i-1}] \leq 0. \tag{57}$$

We then define

$$Z_i = w_i - k_i/2 - d_i - \lambda h_i + (1-p_r)/24, \quad S_0 = 0, \quad S_j = \sum_{i=1}^{j} Z_i. \tag{58}$$

Since $\mathbb{E}[Z_i|\mathcal{T}_{i-1}] \leq 0$, the process $\{S_j\}_{j=0}^{K}$ is a supermartingale. The branch events are mutually exclusive, so the outcomes are in an interval of length

$$\gamma = 1/2 + \max(1, \lambda). \tag{59}$$

Using the Azuma-Hoeffding inequality gives, for every $u \geq 0$,

$$\Pr[S_K \geq u] \leq \exp\left(-\frac{2u^2}{K\gamma^2}\right). \tag{60}$$

For an incorrect computation outcome bit, we need a wrong majority (or unlucky tie-break bit) $W_L(K) = 1 - f(x)$, and the following must be true

$$E_{\text{test}} = \sum_{i=1}^{K} d_i = 0, \tag{61}$$

$$\lambda E_{\text{rigid}} = \lambda \sum_{i=1}^{K} h_i \leq \frac{(1-p_r)K}{96}, \tag{62}$$

$$\sum_{i=1}^{K} w_i \geq \frac{1}{2} \sum_{i=1}^{K} k_i. \tag{63}$$

Therefore

$$\sum_{i=1}^{K} Z_i \geq \frac{(1-p_r)K}{32}, \tag{64}$$

Using (60) with $u = (1-p_r)K/32$ gives us a soundness bound of

$$\Pr[W_L(K) = 1 - f(x)] \leq e^{-(1-p_r)^2 K/512\gamma^2}. \tag{65}$$

□

The value of γ depends on λ which depends on the not-optimized choice of R_0 and p_r, which we leave as an extension for future work. Also note that our sequential Leash' wrapper no longer needs the Q/Q' distinction that CGJV used as computation round statistics determine the bit in $\{0, 1\}$, while test round statistics alone determine whether V outputs err which we need for our later independence requirement.

3.3 Ideal and Real Resources

Having established completeness and soundness in our modified Leash' protocol, we use the abstract cryptography framework to show that sequential Leash' constructs a two-Prover ideal resource with simulatable correctness and blind-verifiability, suitable for multiparty composition. The completeness property will be used to trivially show correctness, while soundness will be an input into our proof of verifiability.

The original single-Prover ideal resource $S_{\text{verif}}^{\text{blind}}$ of Definition 4.2 in Ref. [21] outputs either the correct result or an orthogonal error flag and leaks only the permitted information at the dishonest-server interface. We use the following two-Prover classical analogue.

Definition 3.1 (Modified two-Prover ideal resource). *The resource $S_{\text{verif}}^{\text{blind}\prime}$ receives $x \in \mathcal{X}_\ell$ at the Verifier interface. It reveals $\ell_V(x)$ at the PV interface and $\ell_P(x)$ at the PP interface, and receives control bits $a_V, a_P \in \{0, 1\}$. Its Verifier output is*

$$y_{\text{ideal}} = \begin{cases} f(x), & a_V = a_P = 0, \\ \text{err}, & a_V = 1 \text{ or } a_P = 1. \end{cases} \tag{66}$$

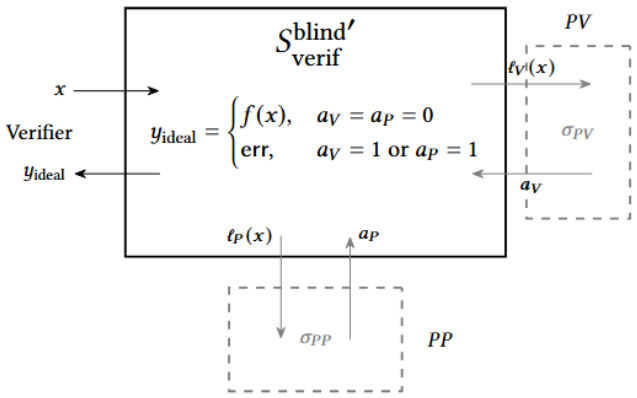


Figure 3: The modified ideal resource $S_{\text{verif}}^{\text{blind}\prime}$ where the Verifier supplies input x to the ideal resource and receives $f(x)$ or err. Separate PV and PP interfaces reveal their permitted leakage and accept delivery-control bits through dashed simulator boxes, additionally requiring one of the two provers to be honest.

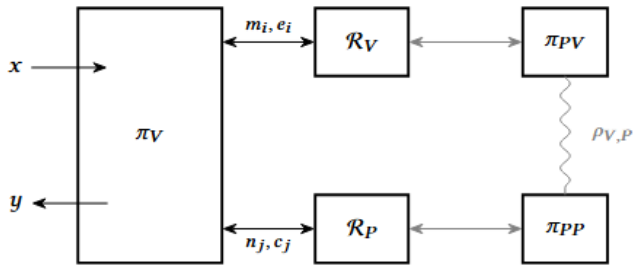


Figure 4: The modified concrete setting, with classical input and output, and a shared quantum state $\rho_{V,P}$ between the two non-communicating Provers. The honest Verifier protocol π_V interacts sequentially with the two Provers through separate classical channels $\mathcal{R}_V$ and $\mathcal{R}_P$. The Provers share an initial entangled state but have no direct communication channel.

The resource never outputs the incorrect valid bit $1 - f(x)$.

The two-Prover interfaces independently control whether the Verifier receives the correct result or an error flag and reveal only $\ell_V(x)$ and $\ell_P(x)$, respectively. For our application, this leakage contains public upper bounds on input size, circuit size, and the number of T gadgets per layer after universal-circuit embedding. Figure 3 also shows the simulators σ_{PV} and σ_{PP} used below.

We represent the sequential Leash' protocol by a concrete resource $\mathcal{R}$, an honest Verifier protocol π_V, and arbitrary Prover protocols π_{PV} and π_{PP}. We prove correctness and blind-verifiability by comparing this construction with $S_{\text{verif}}^{\text{blind}\prime}$ equipped with simulators σ_{PV} and σ_{PP}. For correctness, the distinguisher may access all three interfaces. For blind-verifiability, the threat model exposes the Verifier interface and exactly one Prover interface as we are now assuming at least one honest Prover and we analyze both cases.

Definition 3.2 (Correctness).

$$d(\pi_V \mathcal{R} \pi_{PV}^H \pi_{PP}^H, S_{\text{verif}}^{\text{blind}\prime} \perp_{PV} \perp_{PP}) \leq \epsilon_c, \tag{67}$$

where π_{PV}^H and π_{PP}^H are honest Prover protocols, and ϵ_c is the correctness error.

For honest Provers, the ideal resource always outputs $f(x)$ thus

$$\begin{aligned} d(\mathcal{R}, S_{\text{verif}}^{\text{blind}\prime} \perp) &= \max_x \Pr[\mathcal{R}(x) \neq f(x)] \\ &= \max_x \Big(\Pr[\mathcal{R}(x) = 1 - f(x)] + \Pr[\mathcal{R}(x) = \text{err}] \Big). \end{aligned} \tag{68}$$

Now we use our previously derived sequential Leash' completeness to bound the correctness error.

Theorem 3.3 (Leash' Correctness). *For any input x and honest Verifier protocol π_V^H and honest Prover protocols π_{PV}^H and π_{PP}^H, the sequential Leash' protocol constructs the ideal resource $S_{\text{verif}}^{\text{blind}\prime}$ with correctness error*

$$d(\pi_V \mathcal{R} \pi_{PV}^H \pi_{PP}^H, S_{\text{verif}}^{\text{blind}\prime} \perp_{PV} \perp_{PP}) \leq \epsilon_c, \tag{69}$$

where

$$\epsilon_c = \left(1 - \frac{1-p_r}{3} + \frac{1-p_r}{3} e^{-1/18}\right)^K + e^{-p_r^2 R_0^2 K/2048}. \tag{70}$$

Proof. The distance from the honest ideal resource is just the probability of any output other than $f(x)$ from the real resource, which is $1 - Pr\big[W_L(K) = f(x) | \text{honest}\big]$. The Sequential Leash' Completeness theorem bounds that probability by ϵ_c. □

To prove blind-verifiability, we show that there exist simulators σ_{PV} and σ_{PP} such that, for every distinguisher with access to the Verifier interface and one Prover interface,

$$d(\pi_V \mathcal{R} \pi_{PP}, S_{\text{verif}}^{\text{blind}\prime} \sigma_{PV} \perp_{PP}) \leq \epsilon_V, \tag{71}$$

$$d(\pi_V \mathcal{R} \pi_{PV}, S_{\text{verif}}^{\text{blind}\prime} \perp_{PV} \sigma_{PP}) \leq \epsilon_P, \tag{72}$$

where ϵ_V and ϵ_P are the two Prover-specific security errors. We can then prove local blindness and independent local-verifiability in both cases (malicious-PV honest-PP, and honest-PV malicious-PP) and use the DFPR reduction to prove blind-verifiability. For both cases, we consider a fixed adversarial strategy and permitted leakage value ℓ.

We first show that we can use our sequential Leash' soundness bound for local verifiability for both corruption cases, and then we will go on to prove the local properties needed for DFPR. Let A be the malicious Prover's auxiliary input, B its final private state and transcript, and E the environment. The prescribed setup is prepared inside the protocol map, independently of XAE. Thus $\mathcal{P} : \mathcal{L}(XA) \to \mathcal{L}(YB)$ includes the setup, honest Prover, Verifier, and fixed malicious strategy, with all other honest registers traced out. Inputs have the form

$$\rho = \sum_{x \in \mathcal{X}} p_x |x\rangle\langle x| \otimes \rho_{AE}^x. \tag{73}$$

All channels below act as the identity on E. Their equality is required for every input, including arbitrary correlations between the auxiliary input and reference. As in DFPR let Q copy the valid/error flag from Y into a classical register $\bar{B}$ with basis $\{|\text{ok}\rangle, |\text{err}\rangle\}$. We define ϵ_{ver} for the wrong-output bound in Theorem 3.2. For a fixed classical input x, let ρ_Y be the Verifier's final state and $e = \Pr[Y = \text{err}]$. Since $Y \in \{0, 1, \text{err}\}$,

$$\begin{aligned} &\frac{1}{2} \| \rho_Y - (1-e) |f(x)\rangle\langle f(x)| - e |\text{err}\rangle\langle\text{err}| \|_1 \\ &= \Pr[Y = 1 - f(x)] \leq \epsilon_{\text{ver}}. \end{aligned} \tag{74}$$

This gives local verifiability for each fixed classical input as we can plug in our soundness bound for either case to get

$$\epsilon_{\text{ver}} \leq e^{-(1-p_r)^2 K/512\gamma^2}. \tag{75}$$

We will soon show that e is independent of x for a fixed auxiliary state in the two different corruption cases.

3.3.1 PV Local Blindness.

Lemma 3.1 (Exact local blindness against malicious PV). *Let PP be honest and let PV follow an arbitrary strategy $\mathcal{F}$ on some input $x \in \mathcal{X}_\ell$. There exists a CPTP map $\mathcal{B} : \mathcal{L}(A) \to \mathcal{L}(B)$, independent of x, such that*

$$tr_Y \mathcal{P}(\rho) = \mathcal{B}(tr_X \rho) \tag{76}$$

for every admissible ρ. Hence Leash' is 0-local-blind against malicious PV.

Proof. The questions sent to PV are generated without using x or the PP transcript, they are randomly sampled. Define $\mathcal{B}$ by preparing PV's prescribed setup marginal, generating these questions, interacting with $\mathcal{F}$, and retaining B. This is a CPTP map depending on $\mathcal{F}$ and ℓ, but not on x. Any later interaction with honest PP cannot change this local state. This no-signaling argument also retains E and the Prover's memory across repetitions. Therefore the resulting channel is $\mathcal{B}$ for every fixed x. Applying this channel identity to each classical component of ρ proves (76). □

3.3.2 PV Independent Local Verifiability.

Lemma 3.2 (Independent local-verifiability against malicious PV). *In the setting of Lemma 3.1, every strategy $\mathcal{F}$ has an alternative strategy $\mathcal{F}'$, with an additional register $\bar{B}$, whose induced protocol map $\mathcal{P}'$ satisfies*

$$tr_Y \mathcal{Q}\mathcal{P}(\rho) = tr_Y \mathcal{P}'(\rho). \tag{77}$$

Together with (74), this gives 0-independent ϵ_{ver}-local-verifiability.

Proof. In rigidity and delegation test rounds, the entire honest interaction and its checks that contribute to the verification flag are independent of x. In a computation round, PV's questions are input independent, and PP follows the honest strategy. Thus the only computation-round information affecting the flag is the validity of PV's replies. The remaining input-dependent operations act on honest systems that can be traced out and neither computation counts M_0, M_1 affect the verification flag, which was the motivation for our particular restructuring of the original CGJV protocol. We can use a dummy-input argument to show that the verification flag can be generated locally and independently of the actual input. This argument retains the PV's memory, unused setup systems, and previous verification data, so it applies across all K repetitions.

For any fixed state ψ on AE, any $x, x_0 \in \mathcal{X}_\ell$, and $\rho^x = |x\rangle\langle x| \otimes \psi$, it gives

$$\mathrm{tr}_Y \mathcal{Q}\mathcal{P}(\rho^x) = \mathrm{tr}_Y \mathcal{Q}\mathcal{P}(\rho^{x_0}). \tag{78}$$

The dummy-input execution defines a CPTP map $\mathcal{K} : \mathcal{L}(A) \to \mathcal{L}(B\bar{B})$ by

$$\mathcal{K}(\psi) = \mathrm{tr}_Y \mathcal{Q}\mathcal{P}(|x_0\rangle\langle x_0| \otimes \psi). \tag{79}$$

We then define $\mathcal{F}'$ to discard its actual setup and run this dummy execution internally, preparing both shares of the prescribed setup and simulating the Verifier, honest PP, and $\mathcal{F}$. It uses the auxiliary input A only once, preserving its correlations with E. At the actual interface it returns fixed valid replies, discards the actual interaction record, and outputs the internally generated $B\bar{B}$ all of which can be done locally. This gives

$$\mathrm{tr}_Y \mathcal{P}'(\rho) = \mathcal{K}(\mathrm{tr}_X \rho) = \mathrm{tr}_Y \mathcal{Q}\mathcal{P}(\rho), \tag{80}$$

where the last equality follows from (78) and linearity. This proves (77). □

3.3.3 PP Local Blindness.

Lemma 3.3 (Exact local blindness against malicious PP). *Let PV be honest and let PP follow an arbitrary strategy $\mathcal{G}$. Fix $x \in \mathcal{X}_\ell$, and use $\mathcal{P}$ for the corresponding protocol map. There exists a CPTP map $\mathcal{B} : \mathcal{L}(A) \to \mathcal{L}(B)$, independent of x, such that*

$$tr_Y \mathcal{P}(\rho) = \mathcal{B}(tr_X \rho) \tag{81}$$

for every admissible ρ. Hence Leash' is 0-local-blind against malicious PP.

Proof. In rigidity and delegation test rounds, the interaction is independent of x. In computation rounds, we can apply CGJV's reordering argument from Lemma 4.6 and Eq. (4.3) of [17] which reduces this proof to being essentially the same as Lemma 3.1. The reordered Verifier first interacts with PP using input-independent index sets and uniformly sampled correction bits, and only then performs the input-dependent interaction with honest PV. Let $\tilde{\mathcal{P}}$ denote the protocol with these computation-round interactions reordered. Commutation preserves the reduced channel, including E and PP's memory, so

$$\mathrm{tr}_Y \mathcal{P}(\rho) = \mathrm{tr}_Y \tilde{\mathcal{P}}(\rho). \tag{82}$$

After tracing out the honest computation registers, the postponed operations cannot affect B. The surviving interaction therefore defines a CPTP map $\mathcal{B}$ independent of x that can be applied to each classical component of ρ, proving (81). □

3.3.4 PP Independent Local Verifiability.

Lemma 3.4 (Independent local-verifiability against malicious PP). *In the setting of Lemma 3.3, every strategy $\mathcal{G}$ has an alternative strategy $\mathcal{G}'$, with an additional register $\bar{B}$, whose induced protocol map $\mathcal{P}'$ satisfies*

$$tr_Y \mathcal{Q}\mathcal{P}(\rho) = tr_Y \mathcal{P}'(\rho). \tag{83}$$

Together with (74), this gives 0-independent ϵ_{ver}-local-verifiability.

Proof. We again use the reordered computation interactions from Lemma 3.3, leaving rigidity and delegation test rounds unchanged, in conjunction with the dummy-input argument from Lemma 3.2. The flag in a computation round depends only on transcript validity, which is determined only by PP's replies as PV is honest. Thus the reordering also preserves the joint Prover state and verification flag:

$$\mathrm{tr}_Y \mathcal{Q}\mathcal{P}(\rho) = \mathrm{tr}_Y \mathcal{Q}\tilde{\mathcal{P}}(\rho). \tag{84}$$

In test and rigidity rounds, all operations contributing to the flag are input independent. In a reordered computation round, PP's messages are generated before the input-dependent honest-PV interactions whose operations do not contribute to verification. Tracing out honest-PV's outputs therefore leaves the joint state of $B\bar{B}E$ unchanged. Retaining previous verification data throughout gives, for the same ψ on AE and any $x, x_0 \in \mathcal{X}_\ell$,

$$\mathrm{tr}_Y \mathcal{Q}\mathcal{P}(\rho^x) = \mathrm{tr}_Y \mathcal{Q}\mathcal{P}(\rho^{x_0}), \tag{85}$$

where $\rho^x = |x\rangle\langle x| \otimes \psi$. We can then take a dummy-input x_0 and let $\mathcal{K}$ be the channel that runs this protocol internally on x_0 and returns $B\bar{B}$. Define $\mathcal{G}'$ by the same internal-execution construction as in Lemma 3.2, now simulating honest PV and $\mathcal{G}$. It prepares the prescribed setup, uses A once, and discards the actual interface record. Then

$$\mathrm{tr}_Y \mathcal{P}'(\rho) = \mathcal{K}(\mathrm{tr}_X \rho) = \mathrm{tr}_Y \mathcal{Q}\mathcal{P}(\rho). \tag{86}$$

This proves (83). □

Combining the results of our local security lemmas with DFPR Theorem 5.3 [21] gives us $2(2\sqrt{2\epsilon_{\mathrm{ver}}})$-blind-verifiability for both corruption cases. Using our soundness bound gives us

$$\max\{\epsilon_V, \epsilon_P\} \le 4\sqrt{2}e^{-(1-p_r)^2 K/1024\gamma^2}. \tag{87}$$

4 Multi-Client Protocol

As described by Bartusek [7], a classical MPC replacing a classical Verifier in a delegated computation introduces the input-dependent abort attack. The independent local-verifiability property in the DFPR reduction ensures that the Verifier's error decision reveals nothing beyond the permitted leakage and can safely be handled inside an MPC. For this construction we follow the modular approach of Kashefi and Pappa [31], while adapting it to entirely classical Clients (Verifiers) and a blind-verifiable two-Prover backend using our previously defined threat model in Eq. (7). We start with the classical Client set $C = \{C_1, \ldots, C_M\}$, where Client C_i holds a classical input x_i. Here, $f(\vec{x})$ is the output function of the fixed public computation on the joint inputs given by a universal-circuit embedding, and we assume that all inputs belong to the same length and leakage class $\mathcal{X}_\ell$.

The MPC functionality's final value $Y \in \{0, 1, \mathrm{err}\}$ is delivered to every Client, but not directly to either Prover. The Provers therefore do not receive Y. However, a corrupt Prover learns it only through communicating with a nonempty corrupt-Client set after delivery. In the AC framework, we can combine the interfaces of the colluding corrupt Clients and Prover and treat them as a single adversarial interface. The Prover interfaces are private and disjoint, while Client interfaces receive neither Prover's transcript and cannot relay messages between PV and PP. The setup resource

supplies the prescribed input-independent state. A corrupt Prover may act arbitrarily on its own share of the entangled state, but the coalition cannot choose the setup or replace the share delivered at the honest-Prover interface.

We let V denote the complete classical Verifier of our modified sequential Leash', with converter π_V. For a corrupt coalition $\mathcal{A}$ with combined corrupt client input $\vec{x}_{\mathcal{A}}$, the leakage becomes $(\ell, \vec{x}_{\mathcal{A}})$. For each $\mathcal{A}$, we define a joint simulator $\sigma_{P,\mathcal{A}}$ that receives this leakage. We follow the Kashefi-Pappa approach by defining our ideal functionality as follows.

Definition 4.1 (MPC Verifier functionality). *The resource $\mathcal{F}_{MPC}$ has interfaces for $C_1, \ldots, C_M$, PV, and PP. It records one input per Client and executes the complete Verifier $V(\vec{x})$. Throughout the interaction, it retains the Verifier's randomness, branch choices, Pauli keys, and transcript. It exchanges only the prescribed messages with PV and PP through their private interfaces and, at termination, sends $Y \in \{0, 1, \mathrm{err}\}$ to every Client.*

Protocol Γ consists of Client converters that forward their inputs to $\mathcal{F}_{MPC}$ and return its final output. The resource $\mathcal{E}_{\mathrm{setup}}$ supplies the initial state, and each Prover connects their classical interface with their assigned setup interface. At an honest Prover interface the honest Leash converter is attached to both, whereas a malicious Prover receives both interfaces through the coalition converter. For $\mathcal{A}$, let $\Gamma_{\mathcal{A}}$ denote this entire real hybrid system with precisely the interfaces in $\mathcal{A}$ open and every other interface filtered $\perp_{\bar{I}}$. On the ideal side, $\mathcal{S}_{f,M,t}\sigma_{\mathcal{A}}$ denotes the matching case, with $\sigma_{\mathcal{A}}$ at the coalition interface and the honest filters elsewhere. Here, t specifies the allowed coalition size/threshold.

Definition 4.2 (Ideal Multi-Client Resource). *The ideal resource $\mathcal{S}_{f,M,t}$ has the same Client and Prover interfaces. It fixes one value x_i for each Client (values at corrupt-Client interfaces are chosen adversarially from promised inputs) and computes $b = f(\vec{x})$. If both Prover interfaces are filtered, it sends $Y = b$ to every Client. If exactly one Prover $P \in \{PV, PP\}$ is corrupt, the resource reveals $\ell(\vec{x})$ at that interface, receives a control bit $a_P \in \{0, 1\}$, and sends*

$$Y = \begin{cases} b, & a_P = 0, \\ \mathrm{err}, & a_P = 1. \end{cases} \tag{88}$$

The resource never outputs the incorrect bit $1 - b$. Apart from the corrupt Clients' own inputs, the permitted Prover leakage, and the common Client output, it reveals nothing about the honest Clients' inputs.

Theorem 4.1 (Security of the hybrid composition). *Under the hybrid-model and single-Client assumptions above, for every $\mathcal{A}$ there is a simulator $\sigma_{\mathcal{A}}$ such that*

$$d\big(\Gamma_{\mathcal{A}}, \mathcal{S}_{f,M,t}\sigma_{\mathcal{A}}\big) \le \epsilon_{\mathcal{A}}, \quad \epsilon_{\mathcal{A}} = \begin{cases} \epsilon_c, & \mathcal{A} \subseteq C, \\ \epsilon_V, & PV \in \mathcal{A}, \\ \epsilon_P, & PP \in \mathcal{A}. \end{cases} \tag{89}$$

In particular, the construction error is at most $\epsilon_{\mathcal{A}} \le \max\{\epsilon_c, \epsilon_V, \epsilon_P\}$.

Proof. First suppose that $\mathcal{A} \subseteq C$. The Verifier functionality exposes to corrupt Clients only their own committed inputs and the common output, while both Provers use their honest converters. Routing the single-Client Verifier output to the Client interfaces is a CPTP map, so Theorem 3.3 gives distance at most ϵ_c from the ideal output $f(\vec{x})$. The simulator forwards every corrupt-Client message to the ideal interface and relays that experiment's resulting output.

Now we consider the cases where $PV \in \mathcal{A}$, or $PP \in \mathcal{A}$. By Definition 4.1, the coalition sees $\vec{x}_{\mathcal{A}}$, the messages at PV or PP, and the final output at its corrupt-Client interfaces, but no honest-Client input, internal Verifier state, or transcript from the other Prover. With $\vec{x}_{\mathcal{A}}$ retained, the interactive part of this view is therefore the single-Client modified Leash' execution on $\vec{x}$ with leakage $(\ell, \vec{x}_{\mathcal{A}})$. The corresponding bound in Equation (72) supplies the joint simulator $\sigma_{P,\mathcal{A}}$ with error ϵ_P.

We then define $\sigma_{\mathcal{A}}$ to forward the corrupt Clients' raw submissions to $\mathcal{S}_{f,M,t}$, relay its output at their Client interfaces, and run $\sigma_{P,\mathcal{A}}$ at the open Prover interface. We connect the single-Client ideal resource to the multi-Client resource by identifying $f(\vec{x})$ and passing the simulator's delivery decision to a_P. Connecting the retained register $\vec{x}_{\mathcal{A}}$ and routing each experiment's own Verifier output to the appropriate Client interfaces are the same CPTP post-processing in the real and ideal experiments. Contractivity of trace distance therefore gives the distance ϵ_P. □

4.1 MPC Composition

Corollary 4.1 (MPC Composition). *Suppose a classical real/concrete MPC protocol μ statistically realizes $\mathcal{F}_{MPC}$ with error ϵ_{MPC} under the same threshold t. More precisely, for every $\mathcal{A}$ there is an MPC simulator $\tau_{\mathcal{A}}$ such that*

$$d\big(\mu_{\mathcal{A}}, \mathcal{F}_{MPC}\tau_{\mathcal{A}}\big) \le \epsilon_{MPC}, \tag{90}$$

where $\mu_{\mathcal{A}}$ has the coalition interfaces open and is secure against an arbitrary quantum environment controlling the open Prover interface jointly with the corrupt Clients. Let $\Gamma^{\mu}_{\mathcal{A}}$ be the concrete system obtained by replacing $\mathcal{F}_{MPC}$ with $\mu_{\mathcal{A}}$. AC sequential composition gives a composed simulator $\hat{\sigma}_{\mathcal{A}}$ satisfying

$$d\big(\Gamma^{\mu}_{\mathcal{A}}, \mathcal{S}_{f,M,t}\hat{\sigma}_{\mathcal{A}}\big) \le \epsilon_{MPC} + \epsilon_{\mathcal{A}}. \tag{91}$$

Thus replacing the ideal Verifier functionality adds only the distinguishing error of its classical MPC realization so the total error is at most $\epsilon_{MPC} + \max\{\epsilon_c, \epsilon_V, \epsilon_P\}$.

Proof. Replacing $\mathcal{F}_{MPC}$ by μ changes the composed system by at most ϵ_{MPC} by contractivity under the surrounding protocol. Theorem 4.1 then places the hybrid system within $\epsilon_{\mathcal{A}}$ of the ideal resource. The triangle inequality gives Equation (91), and maximizing over all possible coalitions $\mathcal{A}$ gives the bound $\epsilon_{MPC} + \max\{\epsilon_c, \epsilon_V, \epsilon_P\}$. □

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